Support Vector Machines from a QP Solver and with Kernels

Machine Learning, Homework 5

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Spring 1404 (2025)

Notebook (`svm.ipynb`). The data are Statlog (Landsat) satimage: 36 features, 6 classes. I used a stratified 60/15/25 split and standardised the features.

- *Soft-margin SVM from scratch.* I wrote the dual QP in the standard form $\min \frac{1}{2}\alpha^\top P\alpha + q^\top \alpha$ subject to $G\alpha \le h$ and $A\alpha = b$, and solved it with `cvxopt`. Then I recovered w and b from the support vectors. The binary task uses classes 4 and 5 (the notebook text says 4 and 6; the code uses 4 and 5). With $C = 1$ the linear model reaches accuracy 0.961 and balanced accuracy 0.961 on the test set.
- *RBF kernel, also from scratch.* I picked γ on the validation set by balanced accuracy. The best was $\gamma = 0.001$, which gives test accuracy 0.904, lower than the linear model.
- *Multiclass* `SVC` with an RBF kernel and a grid search over C and γ: best $C = 10$, $\gamma = 0.1$, with accuracy 0.929 and balanced accuracy 0.902. Most errors are between classes 3, 4 and 6 (Figure 1).
- *Kernel comparison* with default settings: linear 0.866, RBF 0.914, polynomial (degree 3) 0.873, sigmoid 0.716 accuracy.
- An optional multiclass SVM trained by sub-gradient descent from scratch reached only 0.46.

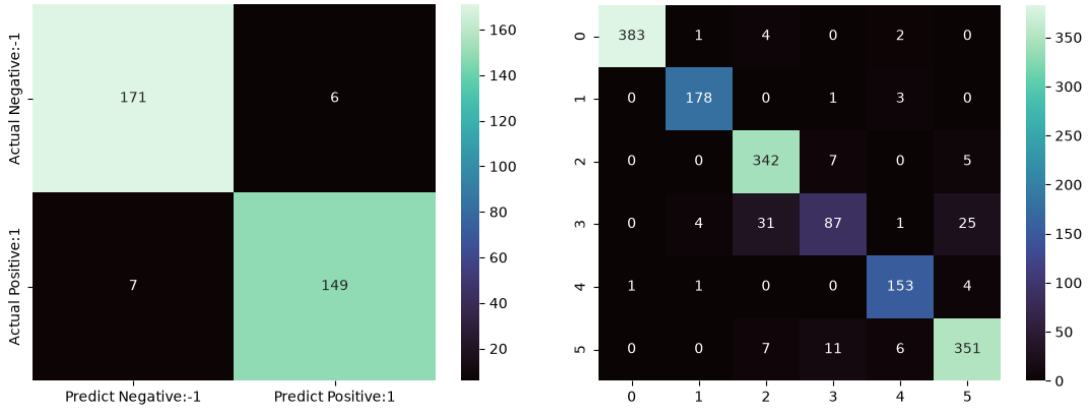


Figure 1: Confusion matrices: from-scratch linear SVM, classes 4 vs. 5 (left); tuned multiclass RBF SVC (right).

Written part.

L_2 soft-margin SVM. The constraint $\xi_i \ge 0$ can be dropped: a negative ξ_i only increases ξ_i^2 and is never needed to satisfy the margin, so the optimum is unchanged. I wrote the Lagrangian and minimised it over w, b and ξ ($w = \sum \alpha_i y_i x_i$, $\sum \alpha_i y_i = 0$, $\xi = \alpha/C$). The dual is $\max_{\alpha \ge 0} \sum \alpha_i - \frac{1}{2}\sum_{ij} \alpha_i \alpha_j y_i y_j (x_i^\top x_j + \delta_{ij}/C)$ with $\sum \alpha_i y_i = 0$.

Two points with a quadratic feature map. For $\phi(x) = [1, \sqrt{2}x, x^2]$, $x_1 = 0$, $x_2 = \sqrt{2}$, the optimal w is parallel to $\phi(x_2) - \phi(x_1) = (0, 2, 2)$. The margin is half the distance between the mapped points, $\sqrt{2}$, so $w = (0, \frac{1}{2}, \frac{1}{2})$.

Kernel closure. Positive scaling, sums and products of valid kernels are valid kernels (the Gram matrices stay positive semidefinite; for the product, use the Schur product theorem).